WORKED SOLUTION

STEP II 1996, Question 3

Pure • Fibonacci Identities by Induction

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s2-q3

The key idea

Everything here follows from the recurrence, so induction is the natural tool. For the product $F_{n+1}F_{n-1} - F_n^2$ the trick is to compare consecutive values: one use of the recurrence turns each value into minus the previous one. For the addition formula, the neat move is to induct on k while letting the hypothesis hold for *every* n at once, so that we may apply it with $n+1$ in place of n.

Solution

The first few values

Using $F_{n+1} = F_n + F_{n-1}$ with $n = 1, 2, 3$:

$$F_2 = F_1 + F_0 = 1 + 0 = 1, \qquad F_3 = F_2 + F_1 = 1 + 1 = 2, \qquad F_4 = F_3 + F_2 = 2 + 1 = 3,$$

as required. Continuing,

$$F_5 = 3 + 2 = 5, \qquad F_6 = 5 + 3 = 8, \qquad F_7 = 8 + 5 = 13.$$

The product $F_{n+1}F_{n-1} - F_n^2$

Write $C_n = F_{n+1}F_{n-1} - F_n^2$ for $n \geq 1$. From the table of values:

n	1	2	3	4	5	6
$F_{n+1}F_{n-1}$	$1 \cdot 0$	$2 \cdot 1$	$3 \cdot 1$	$5 \cdot 2$	$8 \cdot 3$	$13 \cdot 5$
F_n^2	1	1	4	9	25	64
C_n	-1	1	-1	1	-1	1

The values alternate, so we guess

$$F_{n+1}F_{n-1} - F_n^2 = (-1)^n \qquad \text{for all } n \geq 1.$$

Ask yourself How can we get from C_n to C_{n+1} using only the recurrence?

Expand the larger index in C_{n+1}. For $n \geq 1$,

$$\begin{aligned} C_{n+1} &= F_{n+2}F_n - F_{n+1}^2 \\ &= (F_{n+1} + F_n)F_n - F_{n+1}^2 && \text{(recurrence for } F_{n+2}) \\ &= F_n^2 - F_{n+1}(F_{n+1} - F_n) \\ &= F_n^2 - F_{n+1}F_{n-1} && \text{(since } F_{n+1} - F_n = F_{n-1} \text{ for } n \geq 1) \\ &= -C_n. \end{aligned}$$

Now the induction is immediate. The base case is $C_1 = F_2F_0 - F_1^2 = 0 - 1 = -1 = (-1)^1$. If $C_n = (-1)^n$ for some $n \geq 1$, then $C_{n+1} = -C_n = (-1)^{n+1}$. Hence by induction

$$F_{n+1}F_{n-1} - F_n^2 = (-1)^n \qquad \text{for every } n \geq 1.$$

The addition formula

For each positive integer k let $P(k)$ be the statement

$$F_{n+k} = F_kF_{n+1} + F_{k-1}F_n \qquad \text{for } \textit{all} \text{ positive integers } n.$$

Base case $k = 1$. The right-hand side is $F_1F_{n+1} + F_0F_n = 1 \cdot F_{n+1} + 0 = F_{n+1}$, which is the left-hand side. So $P(1)$ holds.

Inductive step. Suppose $P(k)$ holds for some $k \geq 1$, and let n be any positive integer.

Ask yourself The hypothesis is about F_{n+k}, but we need F_{n+k+1}. Which n should we feed it?

Since $P(k)$ holds for every positive integer, we may apply it with $n+1$ in place of n, because $(n+1)+k = n+(k+1)$:

$$\begin{aligned} F_{n+(k+1)} = F_{(n+1)+k} &= F_kF_{n+2} + F_{k-1}F_{n+1} && \text{(by } P(k) \text{ with } n+1) \\ &= F_k(F_{n+1} + F_n) + F_{k-1}F_{n+1} && \text{(recurrence for } F_{n+2}) \\ &= (F_k + F_{k-1})F_{n+1} + F_kF_n \\ &= F_{k+1}F_{n+1} + F_kF_n && \text{(recurrence, as } k \geq 1). \end{aligned}$$

This is exactly $P(k+1)$ for this n, and n was arbitrary. So by induction $P(k)$ holds for every positive integer k, that is,

$$F_{n+k} = F_kF_{n+1} + F_{k-1}F_n \qquad \text{for all positive integers } n, k,$$

as required.

(An alternative is two-step induction on k with n fixed: check $k = 1$ and $k = 2$, then add the formulae for $k-1$ and k. Both routes are fine; the one above needs only one base case.)

Checks

- Cassini with $n = 6$: $F_7F_5 - F_6^2 = 13 \cdot 5 - 64 = 1 = (-1)^6$.
- Addition formula with $n = 3$, $k = 4$: $F_7 = 13$, and $F_4F_4 + F_3F_3 = 9 + 4 = 13$.
- Putting $k = n$ gives $F_{2n} = F_n(F_{n+1} + F_{n-1})$; for $n = 3$ this is $F_6 = 2 \cdot (3 + 1) = 8$.

A computer confirmed the product formula for $1 \leq n < 150$ and the addition formula for all $1 \leq n, k < 90$.

Common slips

- Getting the sign the wrong way round: $C_1 = -1$, so the answer is $(-1)^n$, not $(-1)^{n+1}$.
- In the addition formula, inducting on k with n fixed and then using the hypothesis at $n + 1$: that is only legitimate if the hypothesis is stated for all n.
- Using $F_{n+1} - F_n = F_{n-1}$ without noting it needs $n \geq 1$.

Beyond STEP

Both identities are shadows of one matrix fact: $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}$. Taking determinants gives the product formula (Cassini's identity) at once, and writing $M^{n+k} = M^k M^n$ gives the addition formula. Cassini's identity also shows that consecutive Fibonacci numbers are coprime, and with the addition formula one can prove the striking result $\gcd(F_m, F_n) = F_{\gcd(m,n)}$. Davenport's *The Higher Arithmetic* is a good place to explore this kind of number theory, and Strang's *Introduction to Linear Algebra* treats the Fibonacci numbers through eigenvalues.

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